Upendra Sai Chaturvedula

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Professional Summary

Entry-level AI & Full-Stack Developer with an MCA and B.Sc. in Computer Science. Built full-stack and AI-powered apps, plus a few blockchain projects, in Java, Spring Boot, Python, React, Node.js and Flask, with LLM APIs (Groq). Shipped 8+ end-to-end projects independently. Certified in SQL, Data Analysis and Prompt Engineering, and in Anthropic's Claude and MCP tools. Looking for Software Developer, Full-Stack Developer or AI Engineer roles. Available immediately, open to relocating anywhere in India.

Education

MCA | Avanthi Institute of Engineering & Technology (Affiliated with JNTUH)
2024 to 2026 | Status: Completed
Hyderabad

B.Sc. Computer Science | Vivekavardhani Degree College, Kothagudem (Affiliated with Kakatiya University)
Completed 2024 | Warangal | CGPA: 6.51/10

Intermediate & 10th | Sri Nalanda Jr. College (2019) & Singareni Collieries High School (2017)
CGPA: 6.32/10 & 6.7/10 | Kothagudem

Experience

Academic & Freelance Project Experience
Independent Software & AI Developer **Jul 2025 to Present**

- 8+ applications built and deployed from scratch, covering full-stack, AI and blockchain work. Each has its own database design and REST APIs, with security built in
- Used Claude and Groq in the dev workflow to prototype and debug faster and ship sooner

Technical Skills

Languages: Java (OOP, Java 8+), Python, JavaScript, SQL, PL/SQL, C++

AI & LLM: Groq API, LLM Integration (LLaMA), Claude Code, Gemini, Prompt Engineering, Cloud API Integration

Web Development: React.js (Vite), Node.js, Express.js, Spring Boot, Spring Security, Flask, JSP, Servlets, HTML5, CSS3, REST API Design

Database: MySQL, PostgreSQL, Firestore, CRUD Operations, Database Design

Architecture & Tools: Software Architecture, Fault-Tolerant System Design, Apache Tomcat, Git, GitHub, Docker, Postman, AWS, GCP

Security & Concepts: Counting Bloom Filters (CBF), Cryptographic Verification, Data Security & Encryption, Public Verifiability

Certifications

- SQL and Relational Databases 101
- Data Analysis Using Python
- Prompt Engineering for Everyone
- **Deloitte:** Cyber Security Simulation, Data Analytics Simulation
- **Anthropic:** Claude 101, Claude Code 101, Claude Code in Action, Claude with the Anthropic API, Introduction to MCP, MCP Advanced Topics
- **Udemy:** Complete Prompt Engineering for AI Bootcamp (2026), Complete Full-Stack Web Development Bootcamp

Key Strengths

- **AI-Assisted Efficiency:** Claude, Groq and prompt engineering keep development moving fast
- 8+ projects, built independently and deployed one by one, alongside MCA coursework
- Most of the real problem-solving came from debugging deployment issues and getting APIs to actually talk to each other
- Quick learner. Self-taught new technologies (Groq API, Motoko/ICP, RAG) through independent research
- Cares about writing code that stays clean and scales, and that someone else could maintain without much trouble

Projects

AI Projects

Panda AI: Multi-User AI Assistant Platform
Flask, Python, Groq API, REST APIs
A text chatbot wasn't enough. I wanted something with real web search and file understanding, so I built this instead of relying on a single paid API.

- Multi-user AI assistant with real-time search grounding. Multimodal input covers PDF, image, audio and video files
- Rotates across 3+ Groq keys, then falls back to OpenRouter, so it stays up at zero cost
- Per-user session memory, so several people can use it at once

Universal AI-Powered Resource & Budget ERP System
Java, Groq LLM, REST API
AI can do more than run a chatbot. This explores how it can support real business decisions, like knowing when to reorder stock or flagging a suspicious transaction.

- Multi-domain ERP system. The AI engine gives business insights and handles auto-reordering and fraud detection
- Built the REST API and browser dashboard to support concurrent multi-user sessions across 4+ business modules

Lead Router for Dealers (LRDLR)
JavaScript, Firebase, Node.js, Python (Flask), Groq LLM
Built the lead pipeline first, then added AI on top.

- Routes leads to dealers by pincode prefix, with an In-house fallback. Real-time Firestore dashboard, strict security rules, and a script that cleaned a 40-row legacy CSV (37 kept, 3 rejected)
- 3 AI features via Flask on Render (Groq LLM): smart paste fills the lead form from free text, a dashboard briefing, and fixes for rejected CSV rows. The AI only suggests, a person confirms
- Key kept server-side, with CORS and rate limits. AI output re-checked in code. 17 automated tests, user guide, troubleshooting doc

Java Projects

Secure Cloud Migration & Verifiable Deletion System
Java 8, Counting Bloom Filters, Cryptographic Verification, MySQL, Apache Tomcat
This was my MCA major project at Avanthi Institute, built to solve a real cloud security problem: migrating data safely between providers while guaranteeing permanent, verifiable deletion, without relying on a third party.

- Cloud-to-cloud data transfer and verifiable deletion framework, built on Counting Bloom Filters (CBF), which removes the need for a Trusted Third Party (TTP)
- Added cryptographic confidentiality and integrity validation, plus public verifiability protocols, so malicious server activity can actually be caught

Employee Leave Management System
Java 17, Spring Boot 3, Spring Security, JWT, Spring Data JPA, MySQL/H2
Wanted real hands-on practice with the framework companies keep asking for, so I built a full backend instead of just watching tutorials.

- REST API where employees apply for leave and only their own manager can approve or reject, secured with JWT and role-based access
- Optimistic locking and balance re-checks keep approvals safe. 28 JUnit 5 and Mockito unit tests verify it

Web Development Projects

WaveCoin: Decentralized Token Application
Motoko, Internet Computer (ICP), DFX
Blockchain always felt like a black box, so I built this to see how minting and transfers actually work under the hood, instead of just reading about crypto.

- Custom cryptocurrency logic for minting and transferring tokens on-chain
- Deployed the canisters and built a frontend for real-time balance tracking

Additional Projects

Also built an AI Blog Platform with AI content generation and a RAG chatbot, using React and the Groq LLaMA API, plus an Advanced React Blog Platform. There's an Inventory Management System in Java Swing for real-time stock tracking, and an E-Commerce Console App in Java and MySQL that simulates a full shopping and wallet system. And PeakNFT, an NFT minting platform built with React.js and Solidity/Motoko, using decentralized storage for metadata. Full code for all of these is on GitHub at github.com/Upendrasai1101.